

Question:

Carla is downloading a 200 GB file. Normally she can download 2 GB/minute, but 40% of the way through the download, Windows forces a restart to install updates, which takes 20 minutes. Then Carla has to restart the download from the beginning. How long does it take to download the file?

Reasoning + Answer:

Let's break down the problem step-by-step .

<think>

1 . First , calculate how long it would take to download the full 200 GB without interruptions or restarts :

- $200 \text{ GB} / 2 \text{ GB / minute} = 100 \text{ minutes}$

2 . 40% of the file is $0.4 * 200 \text{ GB} = 80 \text{ GB}$

3 . The first 80 GB would be downloaded at 2 GB per minute , taking :

- $80 \text{ GB} / 2 \text{ GB / minute} = 40 \text{ minutes}$

4 . After the 80 GB , Windows forces a restart . This restart takes 20 minutes .

5 . Now , Carla needs to start from the beginning , so the second 120 GB download starts from scratch :

- $120 \text{ GB} / 2 \text{ GB / minute} = 60 \text{ minutes}$

6 . Total time taken would be :

- $40 \text{ minutes (first 80 GB)} + 20 \text{ minutes (restart)} + 60 \text{ minutes (second 120 GB)}$

7 . $40 + 20 + 60 = 120 \text{ minutes}$

</think>

<answer>

120 </answer>

